

ECO 362: Financial Investments

Midterm Review

Motohiro Yogo

Princeton University

Zero-coupon bond

- A T -year zero-coupon bond with face value \$1.
- Price is related to the yield through

$$P_b(T) = \frac{1}{(1 + Y_b(T))^T}$$

- T : Maturity.
- $P_b(T)$: Price.
- $Y_b(T)$: Yield to maturity.

Annuity

- A T -year annuity with face value \$1.
- Price is related to the yield through

$$P_a(T) = \sum_{t=1}^T \frac{1}{(1 + Y_a(T))^t}$$

- T : Maturity.
- $P_a(T)$: Price.
- $Y_a(T)$: Yield to maturity.

Coupon bond

- A T -year coupon bond with coupon rate c and face value \$1.
- Price is related to the yield through

$$P_c(T) = \sum_{t=1}^T \frac{c}{(1 + Y_c(T))^t} + \frac{1}{(1 + Y_c(T))^T}$$

- T : Maturity.
- $P_c(T)$: Price.
- $Y_c(T)$: Yield to maturity.

No arbitrage relations

- Pricing an annuity relative to zero-coupon bonds.

$$\begin{aligned}P_a(T) &= \sum_{t=1}^T P_b(t) \\ &= \sum_{t=1}^T \frac{1}{(1 + Y_b(t))^t}\end{aligned}$$

- Pricing a coupon bond relative to zero-coupon bonds.

$$\begin{aligned}P_c(T) &= \sum_{t=1}^{T-1} cP_b(t) + (1 + c)P_b(T) \\ &= \sum_{t=1}^{T-1} \frac{c}{(1 + Y_b(t))^t} + \frac{1 + c}{(1 + Y_b(T))^T}\end{aligned}$$

- Pricing a coupon bond relative to an annuity and a zero-coupon bond.

$$\begin{aligned}P_c(T) &= cP_a(T) + P_b(T) \\&= \sum_{t=1}^T cP_b(t) + P_b(T) \\&= \sum_{t=1}^T \frac{c}{(1 + Y_b(t))^t} + \frac{1}{(1 + Y_b(T))^T}\end{aligned}$$

Par yield

- Par yield at maturity T is the coupon rate $c(T)$ at which a T -year coupon bond would trade at face value.

$$\begin{aligned} 1 &= \sum_{t=1}^T \frac{c(T)}{(1 + Y_b(t))^t} + \frac{1}{(1 + Y_b(T))^T} \\ &= \sum_{t=1}^T c(T)P_b(t) + P_b(T) \end{aligned}$$

- Solving for the par yield,

$$c(T) = \frac{1 - P_b(T)}{\sum_{t=1}^T P_b(t)}$$

Forward rate

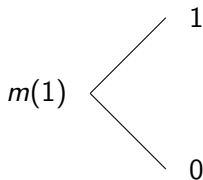
- Today is year 0.
- Forward rate is the interest rate on a loan from year t to T .
- By no arbitrage, related to the zero-coupon yield curve through

$$y_f(t, T) = \frac{1}{T - t} (Ty_b(T) - ty_b(t))$$

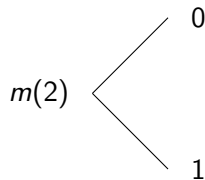
- $y_b(T)$: T -year continuously compounded zero-coupon yield.

Pure state-contingent assets

State 1-contingent asset



State 2-contingent asset



- State prices are the prices of pure state-contingent assets.

Stochastic discount factor

- Define the stochastic discount factor as $M(s) = m(s)/\pi(s)$, which is high for
 - More valuable states: high $m(s)$.
 - Less likely states: low $\pi(s)$.
- The price of any asset i is

$$\begin{aligned} P_i &= \sum_{s=1}^2 m(s)X_i(s) \\ &= \sum_{s=1}^2 \pi(s)M(s)X_i(s) = \mathbb{E}[MX_i] \end{aligned}$$

Present-value formula

- An asset has random payoffs X_1 at date 1, X_2 at date 2, . . .
- M_t is a stochastic discount factor for discounting a random payoff at date t .
- Present-value formula:

$$P = \sum_{t=1}^{\infty} \mathbb{E}[M_t X_t]$$

Two special cases

- Idiosyncratic payoffs (i.e., $\text{Cov}(M_t, X_t) = 0$), such as the insurance contract on baseball players.

$$P = \sum_{t=1}^{\infty} \frac{\mathbb{E}[X_t]}{(1 + Y_b(t))^t}$$

- Constant payoffs like bonds.

$$P = \sum_{t=1}^{\infty} \frac{X_t}{(1 + Y_b(t))^t}$$

Portfolio choice with one risky asset

- Portfolio return is

$$\begin{aligned}R_p &= (1 - w)R_b + wR_s \\ &= R_b + w(R_s - R_b)\end{aligned}$$

where

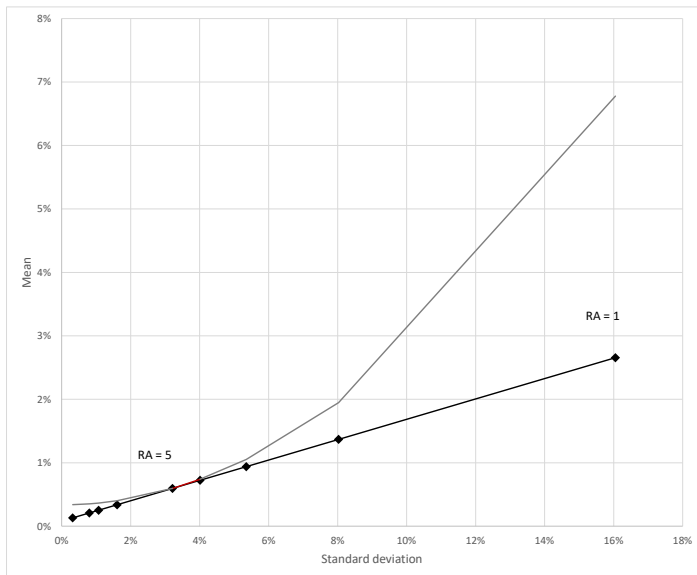
- Riskless rate R_b is constant.
- Stock return R_s has mean μ and variance σ^2 .
- Investor maximizes mean-variance expected utility.

$$\begin{aligned}\max_w \mathbb{E}[U] &= \mathbb{E}[R_p] - \frac{\gamma}{2} \text{Var}(R_p) \\ &= R_b + w(\mu - R_b) - \frac{\gamma}{2} w^2 \sigma^2\end{aligned}$$

- Optimal portfolio is

$$w = \frac{\mu - R_b}{\gamma \sigma^2}$$

Illustration of the optimal portfolio



Portfolio choice with many risky assets

- Portfolio return is

$$\begin{aligned}R_p &= (1 - \mathbf{w}'\mathbf{1})R_b + \mathbf{w}'\mathbf{R} \\ &= R_b + \mathbf{w}'(\mathbf{R} - R_b\mathbf{1})\end{aligned}$$

where

- Riskless rate R_b is constant.
- Vector of stock returns $\mathbf{R}$ has mean μ and covariance matrix Σ .
- Investor maximizes mean-variance expected utility.

$$\begin{aligned}\max_{\mathbf{w}} \mathbb{E}[U] &= \mathbb{E}[R_p] - \frac{\gamma}{2} \text{Var}(R_p) \\ &= R_b + \mathbf{w}'(\mu - R_b\mathbf{1}) - \frac{\gamma}{2} \mathbf{w}'\Sigma\mathbf{w}\end{aligned}$$

- Optimal portfolio is

$$\mathbf{w} = \frac{1}{\gamma} \Sigma^{-1}(\mu - R_b\mathbf{1})$$

Special case when assets are uncorrelated

- In the case of 3 assets,

$$\Sigma = \begin{bmatrix} \text{Var}(R_1) & 0 & 0 \\ 0 & \text{Var}(R_2) & 0 \\ 0 & 0 & \text{Var}(R_3) \end{bmatrix}$$

- The inverse is simple to compute.

$$\Sigma^{-1} = \begin{bmatrix} \text{Var}(R_1)^{-1} & 0 & 0 \\ 0 & \text{Var}(R_2)^{-1} & 0 \\ 0 & 0 & \text{Var}(R_3)^{-1} \end{bmatrix}$$

- Optimal portfolio is

$$\mathbf{w} = \frac{1}{\gamma} \begin{bmatrix} \frac{\mu_1 - R_b}{\text{Var}(R_1)} \\ \frac{\mu_2 - R_b}{\text{Var}(R_2)} \\ \frac{\mu_3 - R_b}{\text{Var}(R_3)} \end{bmatrix}$$

Tangency portfolio

- The tangency portfolio is defined by the weights

$$\hat{\mathbf{w}} = \frac{1}{\hat{\gamma}} \Sigma^{-1} (\mu - R_b \mathbf{1})$$
$$\hat{\gamma} = \mathbf{1}' \Sigma^{-1} (\mu - R_b \mathbf{1})$$

- Weights on the risky assets sum to 1 (i.e., $\hat{\mathbf{w}}' \mathbf{1} = 1$), so the weight on the riskless asset is 0.
- The return on the tangency portfolio is

$$\hat{R}_p = \underbrace{(1 - \hat{\mathbf{w}}' \mathbf{1})}_0 R_b + \hat{\mathbf{w}}' \mathbf{R} = \hat{\mathbf{w}}' \mathbf{R}$$

Portfolio separation theorem

- Every optimal portfolio is a combination of the riskless asset and the tangency portfolio.
- Vector of weights on risky assets is

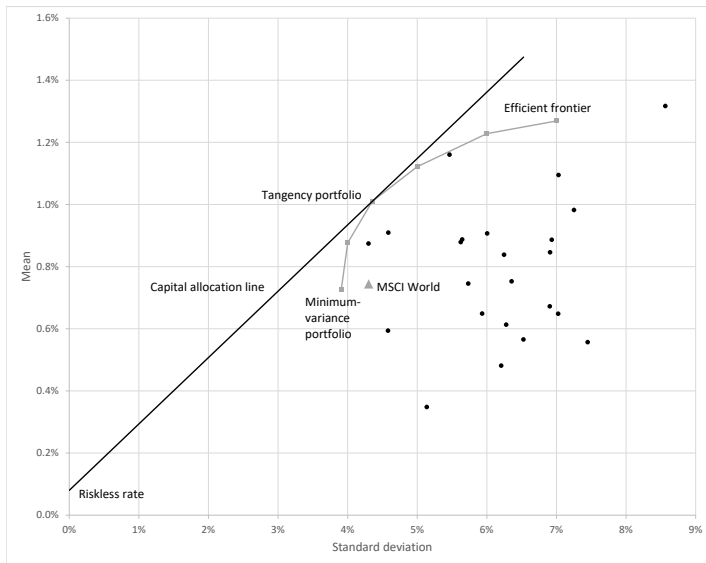
$$\mathbf{w} = \frac{1}{\gamma} \Sigma^{-1} (\mu - R_b \mathbf{1}) = \frac{\hat{\gamma}}{\gamma} \hat{\mathbf{w}}$$

- Weight on the riskless asset is

$$1 - \mathbf{w}' \mathbf{1} = 1 - \frac{\hat{\gamma}}{\gamma} \underbrace{\hat{\mathbf{w}}' \mathbf{1}}_1 = 1 - \frac{\hat{\gamma}}{\gamma}$$

- Given the weights on the tangency portfolio $\hat{\mathbf{w}}$, the optimal portfolio is characterized by one number $\hat{\gamma}/\gamma$.

Illustration of the portfolio separation theorem



Vocabulary test

- A portfolio is **mean-variance efficient** if it has the highest expected return among portfolios with the same variance.
- The **efficient frontier** is a curve that connects all portfolios that are mean-variance efficient.
- The **minimum-variance portfolio** is a portfolio on the efficient frontier that has the lowest variance.
- The **tangency portfolio** is a portfolio on the efficient frontier that has the highest Sharpe ratio.
- The **capital allocation line** connects the tangency portfolio and the riskless rate.

Strategy for preparing

- Do not memorize equations.
- Use the cheat sheet, which should contain the formulas in this review.
- The exam will not ask you to derive the formulas.
- Understand what the formulas mean.
- Be ready to apply the formulas to solve actual problems with numbers.
- All questions are solvable with a hand calculator.